



User's Guide

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SP-X

AMI Yafuflash 7.01 User's Guide

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Chapter 1

1 Introduction

1.1 Purpose

This document is intended to provide information about Yafuflash tool.

1.2 Audience

MegaRAC® SP-X Customers and Customer Support Teams.

1.3 Overview

The Flash tools are command line utility programs used to upgrade the firmware using different mediums like LAN, USB and KCS.

Yet another firmware upgrade flash is a tool used for flashing the BMC. This utility is used for flashing the BMC in three different (Windows, Linux and DOS) environments. The following are three types of mediums used to flash the BMC.

- Network
- USB
- KCS

All three mediums are applicable for Windows and Linux environments. But for DOS environment, only KCS medium is applicable. The medium can be selected as per our requirement.

Note:

YAFU based firmware update using Signed Hashed image is only possible if enough RAM is available to upload the full firmware image before the update starts.

In YAFU firmware upgrade, only YAFU command set is allowed if Enable IPMI Command handling during flashing support is disabled in project configuration.

YAFU flashing process has the following timeout values,

- LAN interface:3600seconds
- USB interface:1800seconds
- KCSinterface:5400seconds

If Secure Boot Support is enabled in the PRJ, YAFUFlash options for Section Based Flashing or Interactive mode will not be used. Hence any feature or options that rely on Section Based Flashing or Interactive mode cannot be used when Secure Boot Support is enabled.

1.4 Yafuflash Features

- Option to display information about current FW and new FW, FW Sections, FW Versions.
- Option to FORCE BootLoader upgrade during full upgrade, and also skips user interaction in Interactive Upgrade mode.
- Option to flash using block by block method.
- Option to flash using OEM specific Netfunction.
- Option to preserve Config Module during full upgrade.
- Option to preserve all the configs by enabling preserve items in web.
- Option to skips user interaction and continues update process.
- Option to skip reselecting the active image.
- Option to flash split image.
- Option to flash particular section Eg. -flash-conf, -flash-[www](#).
- Option to interactive upgrade
- Option to skip the CRC check (Only for DualImageSupport)
- Option to skip the FMH check (Only for DualImageSupport)
- Option to specify the peripheral (Only for DualImageSupport)
- Option to specify the Image to be updated.
- Option to activate peripheral devices.
- Option to give ini file as input.Inifile.
- Option to Flash HPM Image Component wise.
- Option to skip the reboot.
- Option to skip the BMC update with module by module.
- Option to preserve XXXconfiguration. Where XXXfallsin sdr, fru, sel, ipmi, auth, net, ntp, web, snmp, ssh, kvm, syslog, redfish, and rsd.
- Option to replace the Signed Image Key in existing firmware.
- Option to skip flashing modules only if the versions are same.
- Option to use USB Medium.

- Option to use Network Medium.
- Option to use KCS Medium.
- Option to use alternate KCS port (Base address of Data Port, Status Port) when using KCS Medium.
- Option to preserve extlog configuration during firmware flash.
- Option to update image in SD Card.
- Add changes for Windows driver security.
- YafuFlash Show BIOS firmware version.
- Support BIOS preserved parameter functions in Yafuflash.
- Support Redfish preservation option in Yafuflash.
- Needed YAFU tool option to skip preserving UBoot Environmental Variables.
- Option (pweb) to preserve tftp configuration file in Yafuflash.
- Option to erase full SPI.
- Option to Yafu timeout based Image size.
- Option to flash PFR component manager plugin (supports both Intel PFR and AMI PFR).
- Option to Read SPI Data and Dump into file.
- Option to use decommission support.
- Option to use key cancellation support.
- Option to reboot bmc via YAFU. Allowed only in KCS medium.
- Option to initiate firmware update via YAFU with stateless changes.
- Option to specify CPLD device id (Only for CPLD update)
- Option to verify CPLD device interface (Only for CPLD update)
- Option to display YAFU command specific failure error code when flashing failed.
- Option to display-pfupdate and support BIOS component to do parallel firmware update.

1.5 Supported Operating Systems

The following operating systems support Yafuflash tool.

Server OS (In band - USB and KCS mediums)	Client OS (In band and Out of Band)
Windows Server 2022 and 2025	Windows 10 and 11
Ubuntu Server 22.04 and 24.04 LTS	Ubuntu 22.04 and 24.04 LTS
RHEL/CentOS 8 and 9	Fedora 39 and 40
DOS (KCS medium alone)	Debian 11 and 12

Chapter 2

2 YafuFlash Installation

2.1 Installation in Windows

- Download the Yafuflash tool for Windows from git.
- Place and extract the file in the required location.
- This contains Yafuflash.exe, LIBIPMI.dll and amigendrv64.sys.
- Open command prompt and go to path/Yafuflash.
- Run Yafuflash.exe

2.2 Installation in Linux

- OpenSSL is prerequisite for Yafuflash
- Download the Yafuflash tool for Linux from git
- Place and extract the file in the required location
- This contains Yafuflash
- Open terminal and go to path/Yafuflash
- Run ./Yafuflash

Note:

- *Tool can support different OpenSSL versions (currently verified under 1.x and 3.x).*
- *Tool has entries for OpenSSL common symlinks. i.e., "/usr/lib/x86_64-linux-gnu/libcrypto.so" (for Ubuntu) and for direct OpenSSL paths for only supported OS.*
- *By creating symlink to specific openssl version, tool can load different OpenSSL libraries.*

Example:

- *In -s /usr/lib/x86_64-linux-gnu/libcrypto.so.3 /usr/lib/x86_64-linux-gnu/libcrypto.so will create common symlink for Ubuntu and will load corresponding OpenSSL.*
- *In -s /usr/lib64/libcrypto.so.3.0.7 /usr/lib64/libcrypto.so will create common symlink for RHEL. It will load corresponding OpenSSL.*
- *In -s /usr/lib64/libcrypto.so.3.2.2 /usr/lib64/libcrypto.so will create common symlink for FEDORA. It will load corresponding OpenSSL.*

- User can able to list the symlinks using `sudo ls -all /usr/lib64/ | grep libcrypto*`
- By default, first tool will try to initialize OpenSSL using common symlink and if there is no symlink, then it will use direct OpenSSL path listed in the tool entry. If there is no symlink and no direct OpenSSL path, it will fail to initialize openssl library and fail to create LAN session.

2.3 Installation in DOS

- Download the Yafuflash tool for DOS from git
- Place and extract the file in the required location
- This contains Yafuflash.exe and Yafuflash.sym
- Go to the DOS command prompt and navigate to path/Yafuflash
- Run Yafuflash.exe

2.3.1 Driver Unload Manually

If the driver is running already, we can unload it manually by giving following commands in command prompt,

- `sc stop [driver-full-path]`
- `sc delete [driver-full-path]`

Chapter 3

3 Using Yafuflash

3.1 Usage

Yafuflash [OPTION] [FW_IMAGE_FILE]

Perform BMC Flash Update.

-?	Displays the utility usage
-h	Displays the utility usage
-V	Displays the version of the tool
-e	List outs a few examples of the tool
OPTION	
-info	Displays information about current FW and new FW
-msi, -img-section-info	Displays information about current FW Sections
-mi, -img-info	Displays information about current FW Versions
-fb, -force-boot	Option to FORCE BootLoader upgrade during full upgrade. Also, skips user interaction in Interactive Upgrade mode. This option is not allowed with interactive upgrade and -smu option.
-npb, -no-preserve-boot	This option skips preserving uboot envariables. This option cannot used with -smu.
-bu, -block-upgrade	Option to Flash using Block by Block method
-netfn 0xXX	Option to Flash using OEM specific Net function
-pc, -preserve-config	Option to preserve Config Module during full upgrade. If platform supports Dual Image, this option skips user interaction, preserves config and continues

	update process. This option is not allowed with interactive upgrade and -smu option.
-ipc, -ignore-platform-check	If this image is for a different platform, this option skips user interaction and continues update process.
-idi, -ignore-diff-image	If this image differs from the one currently programmed. This option skips user interaction and continues update process
-isi, -ignore-same-image	If this image is same as the one currently programmed, this option skips user interaction and continues update process.
-iml, -ignore-module-location	If module(s) of this image is/are in different locations. This option skips user interaction and continues update process.
-ibv, -ignore-boot-version	If boot loader version is different and -force-boot is not specified, this option skips user interaction and continues update process. The boot loader will be updated.
-iri, -ignore-reselect-image	This option skips reselecting the active image.
-inc, -ignore-non-preserve-config	If the Images of both flashes share the same Configuration area. Not preserving will restore to default factory settings, this option skips it.
-msp, -split-img	Use this option to flash split image
-f-XXX, -flash-XXX	Use this option to flash section where XXX denotes name of the section, example -flash-conf. If it is split image need to give -split-img along with this option.
-q, -quiet	Use the option to show the minimum flash progress details.
-i	Option to interactive upgrade (upgrade only required Modules)
-f, -full	Performs full firmware upgrade with Interactive Upgrade mode. Skips option to select individual module upgrade. This option must be used along with -i(-interactive) option
-sc, -skip-crc	Option to skip the CRC check (Only for Dual Image Support)
-sf, -skip-fmh	Option to skip the FMH check (Only for Dual Image Support)
-d	Option to specify the peripheral • <BIT0> -BMC

	• <BIT1> -BIOS • <BIT2> - CPLD • <BIT4> - ME • <BIT6> - PLDM
-mse, -img-select	Option to specify the Image to be updated. • 0 – Inactive Image • Image 1 • 2 – Image 2
-a, -activate	Option to activate peripheral devices. • <BIT0> -BMC • <BIT1> -BIOS • <BIT2> - CPLD
-ini	Option to give ini file as input. Ini file should be present in the current directory of the Yafuflash executable or in/etc folder • Yafu_SingleImage.ini - For Single Image • Yafu_DualImage.ini - For Dual Image • Yafu_MMCIImage.ini- For MMC Image
-spi, -mmc	Option to Flash HPM Image Component wise • 0 –BOTH • 1-SPI Image • 2 -MMC Image
-nr, -no-reboot	Option to skip the reboot. With online-flash support, if conf/extlog is not preserved, BMC will still reboot
-pXXX, -preserve-XXX	Option to preserve XXX configuration. Where XXX fallsinsdr, fru, sel, ipmi, auth, web, net, ntp, snmp, ssh, kvm, syslog, redfish, and rsd. If the preserve status of another configuration is enabled, then it will ask to confirm that configuration is to be preserved. This option cannot be used with -smu.
-ieo,-ignore-existing-overrides	Clears the existing overrides and preserves only the overrides given in command line if any

-rp,-replace-publickey	Option to replace the Signed Image Key in Existing Firmware
-vcf, -version-cmp-flash	Option to skip flashing modules only if the versions are same by selecting (N/n). Option (Y/y) Selects full firmware upgrade mode.
-non-interactive	This option skips user interaction. This option cannot be used along with 'ignore-diff-image', 'ignore-same-image', '- ignore-module-location' & '- ignore-boot-version' options.
Medium:	
-nw, -ip, -u, -p, -host, -port	Option to use Network Medium 'ip' Option to enter IP, when using Network Medium 'host' Option to enter host name, when using Network Medium 'u' Option to enter Username, when using Network Medium 'p' Option to enter Password, when using Network Medium 'port' Option to enter Port Number
- kcs	Option to use KCS Medium
-kcspport	Option to use alternate KCS port (Base address of DataPort, Status Port) when using KCS Medium. If not givendefault KCS port will be used
FW_IMAGE_FILE:	
fw_image_file	Firmware Image filename
-pe,-preserve-extlog	Option to preserve extlog configuration during firmware flash. This option cannot be used with -smu.
-sd	Option to update image in SD Card

3.2 YAFU v7 new features implemented and its compatibility details

3.2.1 Common features support both LTS12 and LTS13

-showerr,-showerrorcode	Displays YAFU command specific failure error code when flashing failed
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-rbmc, -rebootbmc	Option to reboot bmc via YAFU. Allowed only in KCS medium. Used when flashing failed in the middle.
-pfr	Used to flash the Intel and AMI PFR capsule to the respective device. BIOS, BMC and CPLD devices are supported.
-decommission	This option helps to not provision the PFR server
-cancellation	This option helps to cancel the Key ID for BMC flashing
--active	Capsule will be updated to the Active region
--defer	This option is used to update the uploaded capsule to respective device on BMC reboot
--recovery	Capsule will be updated to both Active and Recovery region
-autospi	Option to enable to stateless flashing if RAM is enabled
-pcwe, -preserve-config-web-enable	Option to preserve all the configs by enabling preserve items in web
-smu, -skip-module-update	Option to skip the BMC update with module by module. This option cannot be used with force boot or preserve config options.
-image-size-dump, -isd	This option takes backup of SPI data with Running Image Size
-dumpspi	This option takes backup of SPI data
-dumpfile	This option takes file name to store SPI flash dump data
-host-spi-select	Option to specify Host SPI (Only for Platguard Support) • 0 – Primary Host SPI • 1- Secondary Host SPI

3.2.2 Specific to LTS13

-cpld-device-id <id>	Option to specify CPLD device id (Only for CPLD update)
-cpld-check-ifc <jtag spi i2c>	Option to verify CPLD device interface (Only for CPLD update)

-pfupdate	Initiate parallel firmware update for selected componenet
-tobc, timeoutbkcmp	Timeout based on Yafu size
-kcswitchport	Option is used for switching secondary kcs port
-kcswitchsmmchannel	Option is used for switching kcs smm channel

Note:

To use link local IPv6 address in Linux OS, user must specify interface name with IP address. Refer network medium example section.

To use YAFU flash (via LAN), it is needed to change default password (due to CA law, we cannot use default password for flashing operations)

CPLD update (-d 0x04) with YAFU v7 will fail because the backend does not implement the CPLD OEM command in LTS12 currently. User may need to use YAFU v6 for CPLD update for LTS12.

Before doing YAFU flash via KCS in Windows version, please disable or delete imbdrv.sys/ipmidrv.sys drivers used for sending ipmiutil KCS commands to avoid YAFU update interruption.

YafuFlash now automatically detects whether the BMC supports Intel PFR or AMI PFR and adjusts its flashing logic accordingly. Capsule validation is performed only for Intel PFR.

Chapter 4

4 Yafuflash Preserve Options

4.1 -preserve-config / -pc

- It is used to preserve full configuration by skipping conf module
- Flash the rom image into BMC
- Modify some of the settings and save it
- Again, flash the rom image via Yafuflash with preserve-config option
- Eg. Yafuflash -nw -ip 10.0.84.57 -u user -p superuser rom.ima -preserve-config
- After flashing, verify the changes are preserved
- Cannot use -pc when existing and flashing conf module sizes/locations are different which may cause BMC corruption.

4.2 -preserve-config-web-enable / -pcwe

- It is used to preserve full configuration by enabling all preserve items in web
- Flash the rom image into BMC
- Modify some of the settings and save it
- Again, flash the rom image via Yafuflash with preserve-config-web-enable option
- Eg. Yafuflash -nw -ip 10.0.84.57 -u user -p superuser rom.ima -preserve-config-web-enable
- After flashing, verify all the configs are preserved

4.3 -preserve-xxx / -pxxx

where xxx can be sdr, fru, sel, ipmi, auth, net, ntp, web, snmp, ssh, kvm, syslog, redfish, and rsd.

- This option is used to preserve individual or partial configurations. If the preserve status of another configuration is enabled, then it will ask to confirm that configuration is to be preserved.
- preserve-xxx option should not be used with preserve-config option.

4.4 -preserve-sdr / -psdr

- It is used to preserve BMC's Sensor Data Records

- It includes the details for all the available sensors. The sensors like Name, Type, Status, Current Reading and Behavior are displayed. The sensor readings are available for Temperature, Fan, Watchdog and Voltage Sensors as well as for supported Discrete Sensors. This page will be refreshed automatically with the latest data retrieved from the database.
- When using -preserve-sdr / -psdr, those data will be preserved.
- This data will be in SDR.dat file and it will be available at/conf/BMC1/project/.
 - Eg. For wolfpas project, SDR.dat will be in /conf/BMC1/wolfpas/

4.5 -preserve-fru / -pfru

- It is used to preserve Field-Replaceable Unit information.
- It includes Basic Information, Chassis Information, Board Information, and Product Information for the BMC's FRU devices.
- When using -preserve-fru / -pfru, those data will be preserved.

4.6 preserve-sel / -psel

- It is used to preserve System Event Logs
- Event Log includes the list of events incurred by the different sensors on this device and their statistics
- Create multiple events using ipmi tool
- When using -preserve-sel / -psel, those data will be preserved.

4.7 -preserve-ipmi / -pipmi

- It is used to preserve IPMI.conf file.
- It will preserve ipmi related configurations
- The IPMI.conf is available at /conf/BMC1/project/.
 - Eg. for wolfpas project SDR.dat will be in/conf/BMC1/wolfpas/

4.8 -preserve-auth / -pauth

- It is used to preserve authentication related settings
- It has settings to enable Active Directory, SSL and sections for setting secret username and password, domain name, domain controller server addresses.
- Modify the configurations and save it.

- When using -preserve-auth / -pauth, those will be preserved.

4.9 -preserve-net / -pnet, -preserve-ntp / -pntp

- It is used to preserve network related settings and /conf/ntp.conf file.
- Modify the network related configurations like enabling/disabling IPv6 support, setting new IP etc., and save it.
- When using -preserve-net / -pnet and -preserve-ntp / -pntp, related settings will be preserved.

4.10 -preserve-web / -pweb

- It is used to preserve web related configurations
- It will be in update firmware.conf
- This file contains the firmware image location details to update firmware configuration.

4.11 -preserve-snmp / -psnmp

- It is used to preserve Simple Network Management Protocol related settings.
- It will preserve /conf/snmp_user.conf/ and /conf/snmp_cfg.conf files.
- snmp_users.conf:
 - This file contains the SNMP user configurations such as username and password encryption mechanism for the specific users.
- snmpcfg.conf:
 - This file contains the SNMP users privilege levels such as read-only user and read-write user

4.12 -preserve-ssh / -pssh

- It is used to preserve SSH related files such as ssh_server_config, ssh_host_dsa_key, ssh_host_dsa_key.pub, ssh_host_rsa_key, ssh_host_rsa_key.pub
- ssh_server_config:
 - This file contains the keyword argument pairs of configurations such as Address family, Accept Env, Allow, users, authorized key files, etc.
- ssh_host_dsa_key, ssh_host_rsa_key:
 - These files contain the private parts of the host keys.
- ssh_host_dsa_key.pub, ssh_host_rsa_key.pub:
 - These files contain the public parts of the host keys.

4.13 -preserve-kvm / -pkvm

- It is used to preserve Keyboard, Video, Mouse and VMedia related settings and it will be available at /conf/kvm.conf, /conf/vmedia.conf.
- Modify the configurations like mouse mode, media settings and save it.
- When using -preserve-kvm / -pkvm, option, those will be preserved.

4.14 -preserve-syslog / -psyslog

- It is used to preserve System Logfiles.
- It has settings for saving local and remote logs, audit logs.
- Modify the configurations and save it.
- When using -preserve-syslog / -psyslog, those will be preserved.
- syslog.conf will be at /conf/syslog.conf.

4.15 -preserve-redfish / -predfish, -preserve-rsd / -prsd

- It is used to preserve redfish database and RSD related configurations.
- Redfish configurations will be in /conf/redfish/

Chapter 5

5 Ethernet Over USB in YAFU

5.1 Overview

SPX 13 supports Ethernet over USB interface. Both Ethernet over USB and IPMI over USB are access through USB interface. Here, the difference is the way of access the USB interface. Reason for recommending, Ethernet over USB is session based and more secure.

Ethernet over USB supports to use any LAN interface (IPMI over LAN support), only difference between IPMI over LAN and Ethernet over USB is that Ethernet over USB can be tested only on the Host operating system.

Note: SPX 13 supports both ethernet over usb and ipmi over usb (-cd option) for USB medium flash.

5.2 Prerequisites

BMC host should be up and running.

Disable power save mode via web UI for testing purpose, so that the USB device/interface will emulate in Host operating system.

5.3 Configuration

Note: Following steps has been tested on Ubuntu 18.04 and RHEL 8.5 host operating system with SPX 13 latest build.

5.3.1 Check BMC Console

After BMC was flashed with image, static USB address will be set as “169.254.0.17” and netmask as “255.255.0.0” by default in BMC for **usb0** interface as shown in below.

Users can check this with “ifconfig usb0” command on the BMC debug console.

```
AMIB2F761807B3A login: sysadmin
Password:
[4333 : 4333 INFO]SERIAL Login from IP:127.0.0.1 user:sysadmin

~ # ifconfig usb0
usb0      Link encap:Ethernet  HWaddr A6:A6:46:F5:8A:76
          inet addr:169.254.0.17  Bcast:169.254.255.255  Mask:255.255.0.0
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0  errors:0  dropped:0  overruns:0  frame:0
          TX packets:0  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```


5.3.2 Disable Power Save Mode in BMC

Launch web UI and Go to **Home -> Settings-> Media Redirection -> VMedia Instance Settings** and disable **Power Save Mode**

Home > Settings > Media Redirection > VMedia Instance Settings

CD/DVD device instances
4

Hard disk instances
4

Remote KVM CD/DVD device instances
4

Remote KVM Hard disk instances
4

☒ Emulate SD Media as USB disk to Host

☐ Power Save Mode

Save

5.3.3 Setting Static IP in Host OS

- After power save mode is disabled via web UI, set static IP and netmask manually as “**169.254.0.18**” and “**255.255.0.0**” respectively in host OS.

Note: You can get the available interfaces in host OS using “ifconfig” command. The following test steps are based on interface name with “usb0”. It may have varied in name i.e., enp0s20f0u1u3c2.

In host OS, give “ifconfig” command.

```

[ami@qa-lab-wp-svr1 ~]$ ifconfig
enol: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 18:c0:4d:24:0a:93 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xf7100000-f7120000

enp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.99.88 netmask 255.255.248.0 broadcast 172.16.103.255
    inet6 fe80::5e85:e8ec:7547:376 prefixlen 64 scopeid 0x20<link>
    ether 68:1c:a2:12:f5:fc txqueuelen 1000 (Ethernet)
    RX packets 93126093 bytes 15248690541 (14.2 GiB)
    RX errors 0 dropped 1433930 overruns 0 frame 0
    TX packets 2057669 bytes 157935377 (150.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 232817 bytes 20490312 (19.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 232817 bytes 20490312 (19.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

usb0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether fe:b6:94:dc:d6:e1 txqueuelen 1000 (Ethernet)
    RX packets 62 bytes 9518 (9.2 KiB)
    RX errors 0 dropped 49 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.122.1 netmask 255.255.255.0 broadcast 192.168.122.255
    ether 52:54:00:e1:07:e4 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

- Set static ip for “USB0” interface as “**169.254.0.18**” and netmask as “**255.255.0.0**” using below command.
- After successful setting, try pinging “**169.254.0.17**” from host OS. Ping should work as below

```
ifconfig usb0 169.254.0.18 netmask 255.255.0.0 up
```

```
ping 169.254.0.17
```

```

[ami@qa-lab-wp-svr1 ~]$ sudo ifconfig usb0 169.254.0.18 netmask 255.255.0.0 up
[sudo] password for ami:
[ami@qa-lab-wp-svr1 ~]$ ping 169.254.0.17
PING 169.254.0.17 (169.254.0.17) 56(84) bytes of data:
64 bytes from 169.254.0.17: icmp_seq=1 ttl=64 time=0.464 ms
64 bytes from 169.254.0.17: icmp_seq=2 ttl=64 time=0.270 ms
64 bytes from 169.254.0.17: icmp_seq=3 ttl=64 time=0.259 ms
64 bytes from 169.254.0.17: icmp_seq=4 ttl=64 time=0.305 ms
64 bytes from 169.254.0.17: icmp_seq=5 ttl=64 time=0.288 ms
64 bytes from 169.254.0.17: icmp_seq=6 ttl=64 time=0.316 ms
64 bytes from 169.254.0.17: icmp_seq=7 ttl=64 time=0.273 ms
^C
--- 169.254.0.17 ping statistics ---

```

- After setting static IP and netmask in host OS, you can check the same using “ifconfig” command.

```

[ami@qa-lab-wp-svr1 Lukman]$ sudo ifconfig usb0 169.254.0.18 netmask 255.255.0.0 up
[sudo] password for ami:
[ami@qa-lab-wp-svr1 Lukman]$ ifconfig
eno1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 18:c0:4d:24:0a:93 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xf7100000-f7120000

enp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.99.88 netmask 255.255.248.0 broadcast 172.16.103.255
    inet6 fe80::5e85:e8ec:7547:376 prefixlen 64 scopeid 0x20<link>
    ether 68:1c:a2:12:f5:fc txqueuelen 1000 (Ethernet)
    RX packets 93794007 bytes 15408383221 (14.3 GiB)
    RX errors 0 dropped 1443830 overruns 0 frame 0
    TX packets 2096814 bytes 160856743 (153.4 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

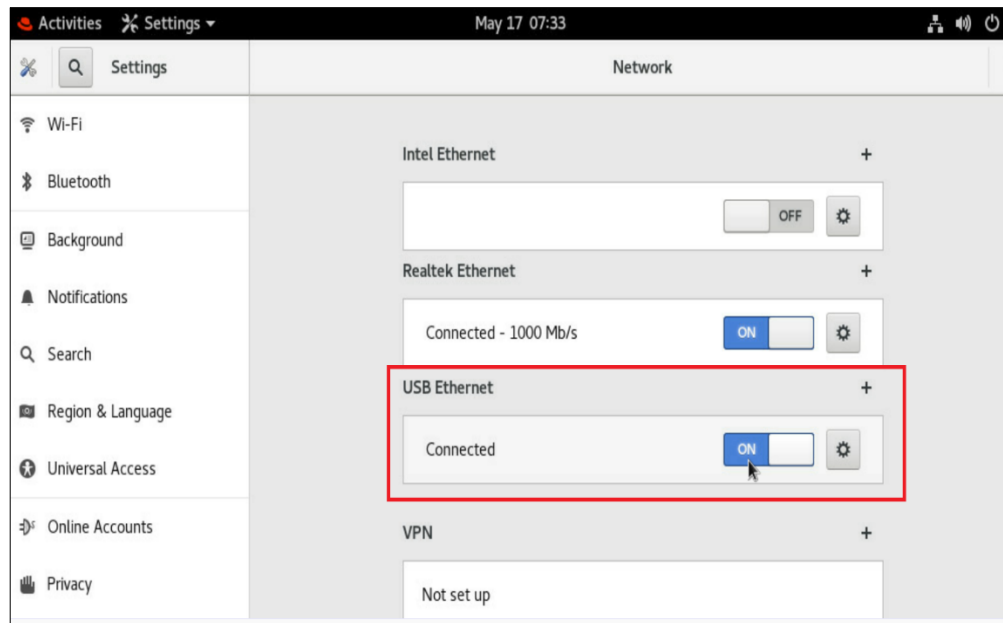
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 232821 bytes 20490520 (19.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 232821 bytes 20490520 (19.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

usb0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 169.254.0.18 netmask 255.255.0.0 broadcast 169.254.255.255
    ether fe:66:dd:45:94:c0 txqueuelen 1000 (Ethernet)
    RX packets 7 bytes 574 (574.0 B)
    RX errors 0 dropped 1 overruns 0 frame 0
    TX packets 9 bytes 1143 (1.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.122.1 netmask 255.255.255.0 broadcast 192.168.122.255
    ether 52:54:00:e1:07:e4 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

- You can check it in host UI also. Go to network setting and check configured interface is connected or not.



- After setting static IP successfully, you can try YAFU flash via ethernet over usb0 from host OS as below.
`./Yafuflash -nw -ip 169.254.0.17 -u admin -p password rom.ima_enc`

```
[ami@qa-lab-wp-svr1 Lukman]$ ./Yafuflash 7.01.0077 -nw -ip 169.254.0.17 -u admin -p superuser LTS-v13_AST2600EV8_FULL1_v20220516_104929_enc.ima_enc
INFO: Yafu INI Configuration File not found... Default options will not be applied...
Creating IPMI session via network with address 169.254.0.17...Done

+-----+
| YAFUFlash - Firmware Upgrade Utility v7.01.0077 |
| Copyright (c) 2020 American Megatrends International, LLC |
+-----+
signed hash length is 256
The Rom Image size = 64 MB
The Current flash size = 64 MB

=====
Firmware Details
=====
+-----+
| ModuleName | Description | Version | ExistingImage |
+-----+
| 1. boot | BootLoader | 13.1.000000 | 13.1.000000 |
| 2. conf | ConfigParams | 13.1.000000 | 13.1.000000 |
| 3. bkupconf |  | 13.1.000000 | 13.1.000000 |
| 4. root | Root | 13.1.000000 | 13.1.000000 |
| 5. osimage | Linux OS | 13.1.000000 | 13.1.000000 |
| 6. www | Web Pages | 13.1.000000 | 13.1.000000 |
| 7. testapps |  | 13.1.000000 | 13.1.000000 |
| 8. bootlogo |  | 13.1.000000 | 13.1.000000 |
| 9. ast2600e |  | 13.1.202017 | 13.1.202017 |
+-----+
Existing Image and Current Image are Same
So, type (Y/y) to do Full Firmware Upgrade or (N/n) to exit
Enter your Option : y

*****
WARNING!
FIRMWARE UPGRADE MUST NOT BE INTERRUPTED ONCE IT IS STARTED.
PLEASE DO NOT USE THIS FLASH TOOL FROM THE REDIRECTION CONSOLE.
*****
Preserving Env Variables... done
Uploading Firmware Image : 100%... done
Skipping [boot] Module ....
Flashing [conf] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [bkupconf] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [root] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [osimage] Module ....
Flashing Firmware Image : 100%... done
```

5.3.4 For USB Port B (USB1)

To use USB port B, it is needed to enable USB port B feature in BMC image and should configure static IP for usb1 as same as usb0.

Steps for setting static IP and netmask are same. Only IP addresses will be differed.

- For USB1, static USB address will be set as “169.254.1.2” and netmask as “255.255.255.0” by default in BMC. User can check this with “ifconfig usb1” on the BMC debug console when USB port B feature is enabled in flashed BMC image.
- After power save mode is disabled via web UI, we should set static IP and netmask manually as “169.254.1.3” and “255.255.255.0” respectively in host OS.
ifconfig usb0 169.254.1.3 netmask 255.255.255.0 up
- After successful setting, try pinging “169.254.1.2” from host OS. Ping should work.
- After setting static IP successfully, you can try YAFU flash via ethernet over usb1 from host OS using below command
.Yafuflash -nw -ip 169.254.1.2 -u admin -p password rom.ima_enc

Chapter 6

6 Network Medium Examples

Eg1: `Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -info`

Description: This command works with network medium using the ip 10.0.84.132, which displays the details of both existing firmware and new firmware.

Eg2: `Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima`

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware.

Eg3: `Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -force-boot`

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade.

Eg4: `Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -preserve-config`

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with preserve config params.

Eg5: `Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -force-boot -preserve-config`

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade and preserve config params.

Eg6: `Yafuflash -nw -host spxbmc -force-boot -preserve-config rom.ima`

Description: This command works with network medium using the host name spxbmc, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade and preserve config params.

Eg7: `Yafuflash -nw -ip 2000::2005 -force-boot rom.ima`

Description: This command works with network medium using the ipv6 address 2000::2005, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade.

Eg8: Yafuflash -nw -ip 10.0.84.132 rom.ima -i

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima using interactive upgrade mode and user will be prompt to select the number of modules and module names to upgrade.

Eg9: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-section-info

Description: This command works with network medium using the ip 10.0.84.132, which displays the details of existing firmware.

Eg10: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-info

Description: This command works with network medium using the ip 10.0.84.132, which displays the details of existing firmware Version.

Eg11: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima --preserve-redfish --preserve-rsd

Description: This command works with network medium using the ip 10.0.84.132, which preserves redfish and rsd related configurations if BMC firmware has that support.

Eg12: Yafuflash -nw -ip 10.0.84.132 rom.ima -preserve-sdr

Description: This command works with network medium using the ip 10.0.84.132, which will ask for other configurations which are already set to be preserved to preserve or not and after that it will start to flash the new rom.ima to the existing firmware with preserving SDR as well as selected configurations.

Eg13: Yafuflash -nw -ip 10.0.84.132 rom.ima -preserve-snmp -preserve-ntp

Description: This command works with network medium using the ip 10.0.84.132, which will ask for other configurations which are already set to be preserved to preserve or not and after that it will start to flash the new rom.ima to the existing firmware with preserving SNMP and NTP as well as selected configurations.

Eg14: Yafuflash -nw -ip 10.0.84.132 rom.ima -preserve-fru -ignore-existing-overrides

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware with preserving FRU configurations only.

Eg15: Yafuflash -nw -ip 10.0.84.132 rom.ima -preserve-fru -preserve-snmp -ignore-existing-overrides

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware with preserving FRU and SNMP configurations only.

Eg16: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -ignore-reselect-image

Description: Yafuflash start full firmware upgrade with default active image. In this it skips the reselecting active image used to flash.

Eg17: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -timeoutbckcmp

Description: Yafuflash will start based on image size with network medium by setting yafu timer with certain values.

Eg18: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -ignore-non-preserve-config

Description: Yafuflash start full firmware upgrade, If the Images of both flash share the same Configuration area. Not preserving will restore to default factory settings, this option skips it

Eg19: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-select 0 rom.ima

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware by selecting the active image to be flashed.

Eg20: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-select 1 rom.ima

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware by selecting the first image to be flashed.

Eg21: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-select 2 rom.ima

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware by selecting the second image to be flashed.

Eg22: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -img-select 3 rom.ima

Description: This command works with network medium using the ip 10.0.84.132, which starts to flash the new rom.ima to the existing firmware by selecting both the images to be flashed.

Eg23: Yafuflash -nw -ip 10.0.84.132 rom.ima -quite

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima with minimum progress details.

Eg24: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser -split-img boot.ima

Description: This command works with network medium to flash the boot split image.

Eg25: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser -split-img root.ima

Description: This command works with network medium to flash the root split image.

Eg26: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser rom.ima -flash-root -flash-conf

Description: This command works with network medium to flash root and conf section from rom.ima file. -flash-<xxx>, where xxx specifies the modules in rom.ima.

Eg27: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser boot.ima -split-img -flash-boot

Description: This command works with network medium to flash root from boot.ima split image. - flash-<xxx>, where xxx specifies the modules in boot.ima.

Eg28: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser root.ima -split-img -flash-www - flash- os image

Description: This command works with network medium to flash www and os image from root.ima split image. - flash-<xxx>, where xxx specifies the modules in root.ima.

Eg29: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser rom.ima -preserve-extlog

Description: This command works with network medium to preserve extended log configuration.

Eg30: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser root.ima -split-img -preserve-extlog

Description: This command works with network medium to preserve extended log configuration from split image.

Eg31: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser root.ima -d 1 rom.ima

Description: This command works with network medium to flash the image on specific peripheral device.

Eg32: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser root.ima -d 1 root.ima -split-img

Description: This command works with network medium to flash the split image on specific peripheral device.

Eg33: Yafuflash -nw-ip 10.0.84.132 -u admin -p superuser -bu root.ima.

Description: This command works with network medium to flash the image on specific peripheral device by block upgrade.

Eg34: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -netfn 0x36

Description: This command works with network medium to flash the image using 0x36 as AMI OEM Net Function instead of default AMI OEM Netfn 0x32.

Eg35: Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima -fe

Description: This command works with network medium and it erases full SPI.

Eg36: Yafuflash -nw -ip 10.0.84.132 -u admin -p admin public.pem --replace-publickey

Description: This command works with network medium and replaces the public key in firmware.

Eg37: Yafuflash -nw -ip 10.0.84.132 -u admin -p admin -pfr pfr.ima --defer --active -pc

Description: This command is to update the PFR image on Active region with preserving configurations

Eg38: Yafuflash -nw -ip 10.0.84.132 -u admin -p admin rom.ima --dumpsbi 0x1000,0x2000
--dumpfile dumpfile.bin

Description: This command is to Read SPI data from offset 0x1000 with Size of 0x2000 and dump into dumpfile.bin file

Eg39: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin rom.ima --image-size-dump -dumpfile dumpfile.bin

Description: This command is to read SPI data from offset 0x0 with Size of running Firmware and dump into dumpfile.bin file

Eg40: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin -pfr --active bmc_signed_cap.bin

Description: This command is to flash the PFR capsule to the respective device. When using the option "--active", the capsule will be updated to the active region.

Eg41: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin -pfr --recovery bmc_signed_cap.bin

Description: This command is to flash the PFR capsule to the respective device. When using the option "--recovery", the capsule will be updated to both active and recovery regions.

Eg42: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin -pfr --defer bmc_signed_cap.bin

Description: This command is to flash the PFR capsule to the respective device. The option "--defer" will be used to update the upload capsule to respective device on reset.

Eg43: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin -cancellation bmc_signed_cap.bin

Description: This command is to cancel the key ID for the BMC flashing

Eg44: Yafuflash -nw -ip 10.0.84.19 -u admin -p admin -decommission bmc_signed_cap.bin

Description: This command is to not provision the PFR server.

Eg45: Yafuflash -nw -ip 10.0.124.41 -u admin -p admin123 -d 2 -host-spi-select 0 Bios.bin

Description: This command is to flash the BIOS image in Primary Host SPI. The option “-host-spi-select” will be used only for Platguard support.

Eg46: ./Yafuflash.exe -nw -ip 10.0.124.72 -U admin -P admin123 rom.ima_enc -autospi

Description: This command is to flash BMC via stateless if RAM available.

Eg47: ./Yafuflash -nw -ip fe80::a856:18ff:fe68:999e%eno1 -u admin -p superuser LTS-v13_AST2600EVB_FULLL1_v20230512_113818_enc.ima_enc

Note: To use link local IPv6 Linux OS, user must specify interface name with IP.

Eg48: ./Yafuflash.exe -nw -ip 10.0.124.72 -U admin -P admin123 -d 2 enc_bios_32.bin -pfupdate

Description: This command is to flash BIOS if RAM is available.

Note: To use link local IPv6 Linux OS, user must specify interface name with IP.

Chapter 7

7 USB Medium Examples

7.1 IPMI over USB

Eg1: Yafuflash -cd rom.ima -info

Description: This command works with USB medium which displays the details of both existing firmware and new Firmware.

Eg2: Yafuflash -cd rom.ima

Description: This command works with USB medium which start to flash the new rom.ima to the existing firmware.

Eg3: Yafuflash -cd rom.ima -force-boot

Description: This command works with USB medium which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade.

Eg4: Yafuflash -cd rom.ima -preserve-config

Description: This command works with USB medium which start to flash the new rom.ima to the existing firmware with preserving config params.

Eg5: Yafuflash -cd rom.ima -force-boot -preserve-config

Description: This command works with USB medium which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade and preserving config params.

Eg6: Yafuflash -cd rom.ima -i

Description: This command works with USB medium, which start to flash the new rom.ima using interactive upgrade mode and user, will be prompt to select the number of modules and module names to upgrade.

Eg7: Yafuflash -cd -img-section-info

Description: This command works with USB medium which displays the details of Existing Firmware.

Eg8: Yafuflash -cd -img-info

Description: This command works with USB medium which displays the details of Existing Firmware Version.

Eg9: Yafuflash -cd rom.ima --preserve-redfish --preserve-rsd

Description: This command works with USB medium which preserves redfish and rsd related configurations if BMC firmware has that support.

Eg10: Yafuflash -cd rom.ima -preserve-sel -preserve-ipmi

Description: This command works with USB medium, which will ask for other configurations which are already set to be preserved to preserve or not and after that it will start to flash the new rom.ima to the existing firmware with preserving SEL and IPMI as well as selected configurations.

Eg11: Yafuflash -cd rom.ima -preserve-sel -ignore-existing-overrides

Description: This command works with USB medium, which start to flash the new rom.ima to the existing firmware with preserving FRU configurations only.

Eg12: Yafuflash -cd rom.ima -ignore-reselect-image

Description: Yafuflash start full firmware upgrade with default active image. In this it skips the reselecting active image used to flash.

Eg13: Yafuflash -cd rom.ima -ignore-non-preserve-config

Description: Yafuflash start full firmware upgrade, if the Images of both flashing share the same Configuration area. Not preserving will restore to default factory settings, this option skips it.

Eg14: Yafuflash -cd -img-select 0 rom.ima

Description: This command works with USB medium, which starts to flash the new rom.ima to the existing firmware by selecting the active image to be flashed.

Eg15: Yafuflash -cd -img-select 1 rom.ima

Description: This command works with USB medium, which starts to flash the new rom.ima to the existing firmware by selecting the first image to be flashed.

Eg16: Yafuflash -cd -img-select 2 rom.ima

Description: This command works with USB medium, which starts to flash the new rom.ima to the existing firmware by selecting the second image to be flashed.

Eg17: Yafuflash -cd -img-select 3 rom.ima

Description: This command works with USB medium, which starts to flash the new rom.ima to the existing firmware by selecting both the images to be flashed.

Eg18: Yafuflash -cd rom.ima -quite

Description: This command works with USB medium, which start to flash the new rom.ima with minimum progress details.

Eg19: Yafuflash -cd -split-img boot.ima

Description: This command works with USB medium to flash the boot split image.

Eg20: Yafuflash -cd -split-img root.ima

Description: This command works with USB medium to flash the root split image.

Eg21: Yafuflash -cd rom.ima -flash-root -flash-conf

Description: This command works with USB medium to flash root and conf section from rom.ima file. -flash-<xxx>, where xxx specifies the modules in rom.ima.

Eg22: Yafuflash -cd boot.ima -split-img -flash-boot

Description: This command works with USB medium to flash root from boot.ima split image. -flash-<xxx>, where xxx specifies the modules in boot.ima.

Eg23: Yafuflash -cd root.ima -split-img -flash-www -flash-osimage

Description: This command works with USB medium to flash www and os image from root.ima split image. -flash-<xxx>, where xxx specifies the modules in root.ima.

Eg24: Yafuflash -cd rom.ima -preserve-extlog

Description: This command works with USB medium to preserve extended log configuration.

Eg25: Yafuflash -cd root.ima -split-img -preserve-extlog

Description: This command works with USB medium to preserve extended log configuration from split image.

Eg26: Yafuflash -cd root.ima -d 1 rom.ima

Description: This command works with USB medium to flash the image on specific peripheral device.

Eg27: Yafuflash -cd root.ima -d 1 root.ima -split-img

Description: This command works with USB medium to flash the split image on specific peripheral device.

Eg28: Yafuflash -cd rom.ima -netfn 0x36

Description: This command works with USB medium to flash the image using 0x36 as AMI OEM Net Function instead of default AMI OEM Netfn 0x32.

Eg29: Yafuflash -cd rom.ima --fe

Description: This command works with USB medium and it erases full SPI.

Eg30: Yafuflash --cd public.pem --replace-publickey

Description: This command works with USB medium, which replaces the public key in Existing Firmware.

Eg31: Yafuflash -cd rom.ima -timeoutbckcmp

Description: Yafuflash will start based on image size with cd medium by setting yafu timer with certain values

Eg32: Yafuflash -cd -pfr --active bmc_signed_cap.bin

Description: This command works with USB medium to flash the PFR capsule to the respective device. When using the option "--active", the capsule will be updated to the active region.

Eg33: Yafuflash -cd -pfr --recovery bmc_signed_cap.bin

Description: This command works with USB medium to flash the PFR capsule to the respective device. When using the option "--recovery", the capsule will be updated to both active and recovery regions.

Eg34: Yafuflash -cd -pfr --defer bmc_signed_cap.bin

Description: This command works with USB medium to flash the PFR capsule to the respective device. The option "--defer" will be used to update the upload capsule to respective device on reset.

Eg35: Yafuflash -cd -cancellation bmc_signed_cap.bin

Description: This command works with USB medium to cancel the key ID for the BMC flashing.

Eg36: Yafuflash -cd -decommission bmc_signed_cap.bin

Description: This command works with USB medium to un provision the PFR server.

7.2 SPX-13 - Ethernet Over USB

Eg1: Yafuflash -nw -ip 169.254.0.17 -u admin -p superuser rom.ima -info

Description: This command works with network medium using the ip 10.0.84.132, which displays the details of both existing firmware and new firmware.

Eg2: Yafuflash -nw -ip 169.254.0.17 -u admin -p superuser rom.ima

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware.

Eg3: Yafuflash -nw -ip 169.254.0.17 -u admin -p superuser rom.ima -force-boot

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade.

Eg4: Yafuflash -nw -ip 169.254.0.17 -u admin -p superuser rom.ima -preserve-config

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with preserve config params.

Eg5: Yafuflash -nw -ip 169.254.0.17 -u admin -p superuser rom.ima -force-boot -preserve-config

Description: This command works with network medium using the ip 10.0.84.132, which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade and preserve config params.

Note:

- For USB0, static USB address will be set as “**169.254.0.17**” and netmask as “**255.255.255.0**” by default in BMC.
- For USB1, static USB address will be set as “**169.254.1.2**” and netmask as “**255.255.255.0**” by default in BMC.

Chapter 8

8 KCS Medium Examples

Eg1: Yafuflash -kcs rom.ima -info

Description: This command works with KCS medium which displays the details of both Existing Firmware and new firmware

Eg2: Yafuflash -kcs rom.ima

Description: This command works with KCS medium which start to flash the new rom.ima to the existing firmware.

Eg3: Yafuflash -kcs rom.ima -force-boot

Description: This command works with KCS medium which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade.

Eg4: Yafuflash -kcs rom.ima -preserve-config

Description: This command works with KCS medium which start to flash the new rom.ima to the existing firmware with preserving config params.

Eg5: Yafuflash -kcs rom.ima -force-boot -preserve-config

Description: This command works with KCS medium which start to flash the new rom.ima to the existing firmware with FORCE BootLoader Upgrade and preserving config params.

Eg6: Yafuflash -kcs rom.ima -i

Description: This command works with KCS medium, which start to flash the new rom.ima using interactive upgrade mode and user, will be prompt to select the Number of modules and module names to upgrade.

Eg7: Yafuflash -kcs -img-section-info

Description: This command works with KCS medium which displays the details of existing firmware.

Eg8: Yafuflash -kcs -img-info

Description: This command works with KCS medium which displays the details of existing firmware version.

Eg9: Yafuflash -kcs rom.ima -preserve-redfish -preserve-rsd

Description: This command works with KCS medium which preserves redfish and rsd related configurations if BMC firmware has that support

Eg10: Yafuflash -kcs rom.ima -preserve-sel -preserve-ipmi

Description: This command works with KCS medium, which will ask for other configurations which are already set to be preserved to preserve or not and after that it will start to flash the new rom.ima to the existing firmware with preserving SEL and IPMI as well as selected configurations.

Eg11: Yafuflash -kcs rom.ima -preserve-sel -ignore-existing-overrides

Description: This command works with KCS medium, which start to flash the new rom.ima to the existing firmware with preserving FRU configurations only.

Eg12: Yafuflash -kcs rom.ima -ignore-reselect-image

Description: Yafuflash start full firmware upgrade with default active image. In this it skips the reselecting active image used to flash.

Eg13: Yafuflash -kcs rom.ima -ignore-non-preserve-config

Description: Yafuflash start full firmware upgrade, if the Images of both flashing share the same Configuration area. Not preserving will restore to default factory settings, this option skips it

Eg14: Yafuflash -kcs -img-select 0 rom.ima

Description: This command starts to flash the new rom.ima to the existing firmware by selecting the active image to be flashed.

Eg15: Yafuflash -kcs -img-select 1 rom.ima

Description: This command starts to flash the new rom.ima to the existing firmware by selecting the first image to be flashed.

Eg16: Yafuflash -kcs -img-select 2 rom.ima

Description: This command starts to flash the new rom.ima to the existing firmware by selecting the second image to be flashed.

Eg17: Yafuflash -kcs -img-select 3 rom.ima

Description: This command starts to flash the new rom.ima to the existing firmware by selecting both the images to be flashed.

Eg18: Yafuflash -kcs rom.ima -quite

Description: This command works with KCS medium, which start to flash the new rom.ima with minimum progress details.

Eg19: Yafuflash -kcs -split-img boot.ima

Description: This command works with KCS medium to flash the boot split image.

Eg20: Yafuflash -kcs -split-img root.ima

Description: This command works with KCS medium to flash the root split image.

Eg21: Yafuflash -kcs rom.ima -flash-root -flash-conf

Description: This command works with KCS medium to flash root and conf section from rom.ima file. -flash-<xxx>, where xxx specifies the modules in rom.ima.

Eg22: Yafuflash -kcs boot.ima -split-img -flash-boot

Description: This command works with KCS medium to flash root from boot.ima split image. -flash-<xxx>, where xxx specifies the modules in boot.ima.

Eg23: Yafuflash -kcs root.ima -split-img -flash-www -flash-osimage

Description: This command works with KCS medium to flash www and osimage from root.ima split image. -flash-<xxx>, where xxx specifies the modules in root.ima.

Eg24: Yafuflash -kcs rom.ima -preserve-extlog

Description: This command works with KCS medium to preserve extended log configuration.

Eg25: Yafuflash -kcs root.ima -split-img -preserve-extlog

Description: This command works with KCS medium to preserve extended log configuration from split image.

Eg26: Yafuflash -kcs root.ima -d 1 rom.ima

Description: This command works with KCS medium to flash the image on specific peripheral device.

Eg27: Yafuflash -kcs root.ima -d 1 root.ima -split-img

Description: This command works with KCS medium to flash the split image on specific peripheral device.

Eg28: Yafuflash -kcs rom.ima -netfn 0x36

Description: This command works with KCS medium to flash the image using 0x36 as AMI OEM Net Function instead of default AMI OEM Netfn 0x32.

Eg29: Yafuflash -kcs rom.ima -fe

Description: This Command works with KCS medium and it erases full SPI.

Eg30: Yafuflash -kcs public.pem --replace-publickey

Description: This command works with KCS medium, which replaces the public key in Existing Firmware.

Eg31: Yafuflash -kcs rom.ima -timeoutbckcmp

Description: Yafuflash will start based on image size with kcs medium by setting yafu timer with certain values.

Eg32: Yafuflash -kcs -pfr --active bmc_signed_cap.bin

Description: This command works with KCS medium to flash the PFR capsule to the respective device. When using the option "--active", the capsule will be updated to the active region.

Eg33: Yafuflash -kcs -pfr --recovery bmc_signed_cap.bin

Description: This command works with KCS medium to flash the PFR capsule to the respective device. When using the option "--recovery", the capsule will be updated to both active and recovery regions.

Eg34: Yafuflash -kcs -pfr --defer bmc_signed_cap.bin

Description: This command works with KCS medium to flash the PFR capsule to the respective device. The option "--defer" will be used to update the upload capsule to respective device on reset.

Eg35: Yafuflash -kcs -cancellation bmc_signed_cap.bin

Description: This command works with KCS medium to cancel the key ID for the BMC flashing

Eg36: Yafuflash -kcs -decommission bmc_signed_cap.bin

Description: This command works with KCS medium to un provision the PFR server.

Chapter 9

9 Sample Snapshots

Yafuflash -?

```
D:\BMC\Yafuflash>Yafuflash.exe -?
INFO: Yafu INI Configuration File not found... Default options will not be applied...
Usage: Yafuflash [OPTION] [FW_IMAGE_FILE]
Perform BMC Flash Update
-? Displays the utility usage
-h Displays the utility usage
-V Displays the version of the tool
-e List outs a few examples of the tool
OPTION :
-info Displays information about current FW and new FW.
-msi,-img-section-info Displays information about current FW Sections.
-mi,-img-info Displays information about current FW Versions.
-fb,-force-boot Option to FORCE BootLoader upgrade during full upgrade.
Also, skips user interaction in Interactive Upgrade mode.
This option is not allowed with interactive upgrade option
-npb,-no-preserve-boot This option skips preserving uboot env variables
-bu,-block-upgrade Option to Flash using Block by Block method
-netfn 0xXX Option to Flash using OEM specific Netfunction
-pc,-preserve-config Option to preserve Config Module during full upgrade.
If platform supports Dual Image, this option skips user
interaction, preserves config and continues update process.
This option is not allowed with interactive upgrade option.
-ipc,-ignore-platform-check If this image is for a different platform, this option skips
user interaction and continues update process.
-idi,-ignore-diff-image If this image differs from the one currently programmed, this
option skips user interaction and continues update process.
-isi,-ignore-same-image If this image is same as the one currently programmed, this
option skips user interaction and continues update process.
-impl,-ignore-module-location If module(s) of this image is/are in a different location, this
option skips user interaction and continues update process.
-ibv,-ignore-boot-version If bootloader version is different and -force-boot is not specified,
this option skips user interaction and continues update process.
The bootloader will be updated.
```

Figure 1: Help Contents of Yafuflash tool

Yafuflash -v & Yafuflash -e

```
prakash@prakash-desktop:~/BMC-UTIL/yafuflash/packages/yafuflash-ANY/data/obj/Linux_x86_64$ ./Yafuflash -v
INFO: Yafu INI Configuration File not found... Default options will not be applied...

-----+-----
YAFUFlash - Firmware Upgrade Utility v7.00.0004
Copyright (c) 2020 American Megatrends International, LLC |

prakash@prakash-desktop:~/BMC-UTIL/yafuflash/packages/yafuflash-ANY/data/obj/Linux_x86_64$ ./Yafuflash -e
INFO: Yafu INI Configuration File not found... Default options will not be applied...
EXAMPLES :
NETWORK Medium:
Flash the Firmware with preserving boot section          : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima
Flash the Firmware without preserving boot section       : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima {-force-boot | -fb}
Flash the Firmware with preserving both boot and conf section : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima {-preserve-config | -pc}
Flash the Firmware with preserving conf section only      : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima {-force-boot | -fb} {-preserve-conf}
Flash the Firmware by using BMC Hostname                 : # Yafuflash -nw -host bmchost -u admin -p admin rom.ima {-force-boot | -fb}

Displays the information about the Images                 : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima -info
Flash the ME Firmware Firmware                          : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin -d <BIT16> ME_Img

USB Medium:
Flash the Firmware with USB Medium                      : # Yafuflash -cd rom.ima
Flash the Firmware with preserving conf section         : # Yafuflash -cd rom.ima {-preserve-config | -pc}

KCS Medium:
Flash the Firmware with KCS Medium                      : # Yafuflash -kcs rom.ima
Flash the Firmware with KCS Medium using OEM specific Netfun : # Yafuflash -kcs -netfn 0xxx rom.ima
Flash the Firmware with KCS Medium using alternate KCS port : # Yafuflash -kcs -kcsport 0xxx,0xxx <DataPort,StatusPort> rom.ima

Flash the Firmware with preserving conf section         : # Yafuflash -kcs rom.ima {-preserve-config | -pc}

QUIET Operation:
Full Upgrade Mode                                     : # Yafuflash [MEDIUM] {-full | -f} {-preserve-config | -pc} {-ignore-platform-check | -ipc}
                                                    {-ignore-module-location | -iml} {-ignore-same-image | -isi} {-ignore-boot-version | -ibv}
                                                    [FW_IMAGE_FILE]
Flash Split Image Firmware                             : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin {-split-img | -msp} root.ima
Non Interactive section based upgrade                  : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin rom.ima {-flash-conf | -f-conf} {-flash-wd}
Non Interactive section based upgrade with split image : # Yafuflash -nw -ip xx:xx:xx:xx -u admin -p admin root.ima {-split-img | -msp} {-flash-osima}
Flash the HPM Image using network medium               : # Yafuflash -nw -ip xx.xx.xx.xx -u admin -p admin output.hpm
Flash the HPM Image component using network medium     : # Yafuflash -nw -ip xx.xx.xx.xx -u admin -p admin output.hpm {-spi | -mmc}
Flash the Single Image using INI File                   : # Yafuflash -ini Yafu_SingleImage.ini rom.ima
Flash the Dual Image using INI File                    : # Yafuflash -ini Yafu_DualImage.ini rom.ima
Flash the MMC using INI File                           : # Yafuflash -ini Yafu_MMCImage.ini rom.ima
Interactive Upgrade Mode                               : # Yafuflash [MEDIUM] -i {-ignore-platform-check | -ipc}
                                                    {-ignore-diff-image | -idi} {-full | -f} [FW_IMAGE_FILE]
```

Figure 2: Example Commands of Yafuflash tool

```
Yafuflash -nw -ip 10.0.84.132 -u admin -p superuser rom.ima_enc -info
```

```
prakash@prakash-desktop:~/BMC-UTIL/yafuflash/packages/yafuflash-ANY/data/obj/Linux_x86_64$ ./Yafuflash
INFO: Yafu INI Configuration File not found... Default options will not be applied...

Creating IPMI session via network with address 10.0.84.10...Done

+-----+
| YAFUFlash - Firmware Upgrade Utility v7.00.0004 |
| Copyright (c) 2020 American Megatrends International, LLC |
+-----+
signed hash length is 256

=====
Firmware Details
=====
```

			RomImage	ExistingImage
	ModuleName	Description	Version	Version
1.	boot	BootLoader	12.3.000000	12.3.000000
2.	conf	ConfigParams	12.3.000000	12.3.000000
3.	bkupconf		12.3.000000	12.3.000000
4.	root	Root	12.3.000000	12.3.000000
5.	osimage	Linux OS	12.3.000000	12.3.000000
6.	www	Web Pages	12.3.000000	12.3.000000
7.	lmedia		12.3.000000	12.3.000000
8.	extlog	ExtendedLog	2.1.000000	2.1.000000
9.	testapps		2.4.000000	2.4.000000
10.	bootlogo		1.0.000000	1.0.000000
11.	dre		12.3.000000	12.3.000000
12.	wolfpass		12.3.220035	12.3.220035

```
prakash@prakash-desktop:~/BMC-UTIL/yafuflash/packages/yafuflash-ANY/data/obj/Linux_x86_64$ ^C
```

Figure 3: Information about the image file

Yafuflash.exe -nw -ip 10.0.84.10 -u admin -p superuser rom.ima_enc

```

INFO: Yafu INI Configuration File not found... Default options will not be applied...
Creating IPMI session via network with address 10.0.84.10...Done

+-----+
+  YAFUFlash - Firmware Upgrade Utility v7.00.0004  +
+ Copyright (c) 2020 American Megatrends International, LLC +
+-----+

signed hash length is 256
The Rom Image size = 32 MB
The Current flash size = 32 MB
=====
                        Firmware Details
=====
                        RomImage      ExistingImage
Module Name  Description  Version      Version
1. boot      BootLoader  12.3.000000  12.3.000000
2. conf      ConfigParams 12.3.000000  12.3.000000
3. bkupconf  ConfigParams 12.3.000000  12.3.000000
4. root      Root        12.3.000000  12.3.000000
5. osimage   Linux OS    12.3.000000  12.3.000000
6. www       Web Pages   12.3.000000  12.3.000000
7. testapps  Test Apps   2.4.000000   2.4.000000
8. ast2500e  AST2500E    12.3.220035  12.3.220035
Existing Image and Current Image are Same
Continuing with Full Firmware Update

*****
WARNING!
FIRMWARE UPGRADE MUST NOT BE INTERRUPTED ONCE IT IS STARTED.
PLEASE DO NOT USE THIS FLASH TOOL FROM THE REDIRECTION CONSOLE.
*****
Preserving Env Variables... done
Regenerating Boot Variables
Uploading Firmware Image : 100%... done
Flashing [boot] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Setting Env variables... done
Flashing [conf] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [bkupconf] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [root] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [osimage] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [www] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [testapps] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Flashing [ast2500e] Module ....
Flashing Firmware Image : 100%... done
Verifying Firmware Image : 100%... done
Setting Env variables... done
Resetting the firmware.....

```

Figure 4: BMC Flash Update

```
/Yafuflash.exe -nw -ip 10.0.124.72 -U admin -P admin123 rom.ima_enc -autospi
```

```
./Yafuflash -nw -ip 10.0.124.72 -U admin -P admin123 /home/suresh/code/MST/feb10/image_sign/v7/a11_fp_1.ima_enc -autospi
INFO: Yafu INI Configuration File not found... Default options will not be applied...

Creating IPMI session via network with address 10.0.124.72...Done

+-----+
| YAFUFlash - Firmware Upgrade Utility v7.01.007 |
| Copyright (c) 2020 American Megatrends International, LLC |
+-----+

-----
YAFUFlash -Updating SPI Image stateless
-----

Uploading Firmware Image: 100%... done

Verification: Done
Prepare Device: Done
Flashing: Done
SPI Check: Done
Flashing: Flash completed..
```

Figure 5: BMC Update via stateless